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Focus on Unit 2 (Differential Calculus-I) and Unit 4 (Multiple Integration). Together they account for about 63 marks among all questions printed per paper. Unit 3 (Differential Calculus-II) is safest to skim if you're short on time.

Full subject strategy >

TOP UNITS See all >

Unit 2 - Differential Calculus-I Notes ↗ Important questions ↗

Unit 4 - Multiple Integration Important questions ↗

PAPER ID-311753

Subject Code: BAS103

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM I) THEORY EXAMINATION 2025-26
ENGINEERING MATHEMATICS-I

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	In the matrix equation $\begin{bmatrix} 3 & -1 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$, what are the values of x and y? आव्यूह समीकरण $\begin{bmatrix} 3 & -1 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$ में, x और y की वैल्यू क्या हैं?	1	K2
b.	Find the Rank of the following Matrix A: निम्नलिखित आव्यूह A का रैंक ज्ञात कीजिए: $A = \begin{bmatrix} 2 & -4 & 6 \\ -1 & 2 & -3 \\ 3 & -6 & 9 \end{bmatrix}$.	1	K5
c.	If $y = x^3 \log_e x$, prove that $\frac{d^4y}{dx^4} = \frac{6}{x}$. अगर $y = x^3 \log_e x$, साबित करें $\frac{d^4y}{dx^4} = \frac{6}{x}$.	2	K1
d.	If $u(x, y) = (\sqrt{x} + \sqrt{y})^5$, then find the degree of the function $u(x, y)$. यदि $u(x, y) = (\sqrt{x} + \sqrt{y})^5$, फिर फलन की घात ज्ञात कीजिए।	2	K5
e.	If $x = r\cos\theta$, $y = r\sin\theta$, find $\frac{\partial(r,\theta)}{\partial(x,y)}$. यदि $x = r\cos\theta$, $y = r\sin\theta$, $\frac{\partial(r,\theta)}{\partial(x,y)}$ ज्ञात कीजिए।	3	K3
f.	Evaluate the following integral using the Beta-Gamma function: बीटा-गामा फ़ंक्शन का उपयोग करके निम्नलिखित समाकलन का मूल्यांकन करें: $\int_0^2 x(8 - x^3)^{1/3} dx$.	4	K4
g.	Show that gradient field describing a motion is irrotational. यह सिद्ध कीजिए कि किसी गति का वर्णन करने वाला ग्रेडिएंट क्षेत्र अघूर्णी होता है।	5	K4

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SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

Q no.	Question	CO	Level
a.	Determine the values of λ and μ such that the following system: $2x-5y+2z=8$ $2x+4y+6z=5$ $x+2y+\lambda z=\mu$ has (i) no solution (ii) a unique solution (iii) infinitely many solutions. λ और μ के मान निर्धारित करें जैसे कि निम्न प्रणाली: $2x-5y+2z=8$ $2x+4y+6z=5$ $x+2y+\lambda z=\mu$ (i) कोई समाधान नहीं है (ii)एक अद्वितीय समाधान है (iii) अपरिमित रूप से कई समाधान हैं।	1	K2
b.	Find the nth derivative of $\frac{2x+1}{(2x-1)(2x+3)}$. $\frac{2x+1}{(2x-1)(2x+3)}$ का n^{th} डेरिवेटिव ज्ञात कीजिए।	2	K6
c.	Expand $x^2y+3y-2$ in powers of $(x-1)$ and $(y+2)$ up to 3rd degree using Taylor's series . टेलर की श्रेणी का उपयोग करके $x^2y+3y-2$, $(x-1)$ और $(y+2)$ का तीसरी घात तक विस्तार करें।	3	K3
d.	Change the order of integration and hence evaluate : इंटीग्रेशन का क्रम बदलें और उसका मूल्यांकन करें: $\int_0^a \int_{\sqrt{ax}}^a \frac{y^2 dydx}{\sqrt{y^4-a^2x^2}}$.	4	K4
e.	Find the directional derivative of the divergence of $f(x,y,z)=xy\hat{i}+xy^2\hat{j}+z^2\hat{k}$ at the point (2, 1, 2) in the direction of the outer normal to the sphere $x^2+y^2+z^2=9$. $x^2+y^2+z^2=9$ गोले के बाहरी नॉर्मल की दिशा में पॉइंट (2, 1, 2) पर डाइवर्जेंस $f(x,y,z)=xy\hat{i}+xy^2\hat{j}+z^2\hat{k}$ का डायरेक्शनल डेरिवेटिव पता करें।	5	K5

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q	Question	CO	Level

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no.			
a.	Find the eigen values and eigen vectors of the following Matrix A: नीचे दिए गएआव्यूह A के आइगन वैल्यू औरआइगन वेक्टर पता करें: $A=\begin{bmatrix}6 & -2 & 2\\ -2 & 3 & -1\\ 2 & -1 & 3\end{bmatrix}$.	1	K5
b.	If $X_1=\begin{bmatrix}3\\1\\-4\end{bmatrix}$, $X_2=\begin{bmatrix}2\\2\\-3\end{bmatrix}$, $X_3=\begin{bmatrix}0\\-4\\1\end{bmatrix}$, then show that the vectors X_1, X_2, X_3 are linearly dependent. यदि $X_1=\begin{bmatrix}3\\1\\-4\end{bmatrix}$, $X_2=\begin{bmatrix}2\\2\\-3\end{bmatrix}$, $X_3=\begin{bmatrix}0\\-4\\1\end{bmatrix}$, तो दर्शाइये कि सदिश X_1, X_2, X_3 रैखिकतःआश्रित हैं।	1	K2

4. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	If $z=x^4y^2\sin^{-1}\left(\frac{x}{y}\right)+\log x-\log y$, showthat $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=6x^4y^2\sin^{-1}\left(\frac{x}{y}\right)$. यदि $z=x^4y^2\sin^{-1}\left(\frac{x}{y}\right)+\log x-\log y$, , तो दर्शाइये $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=6x^4y^2\sin^{-1}\left(\frac{x}{y}\right)$.	2	K5
b.	Trace the curve : वक्र का अनुरेखण करें: $y^2(a+x)=x^2(3a-x)$.	2	K1

5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Find the minimum distance from the point $(1,2,0)$ to the cone $z^2=x^2+y^2$. बिंदु $(1,2,0)$ सेशंकु $z^2=x^2+y^2$ तक की न्यूनतम दूरी ज्ञात करें।	3	K3
b.	Find approximate value of $\left[(0.98)^2+(2.01)^2+(1.94)^2\right]^{1/2}$.	3	K5

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	$\left[(0.98)^2+(2.01)^2+(1.94)^2\right]^{1/2}$ का अनुमानित मान ज्ञात कीजिए।		

6. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Prove that the area in the positive quadrant bounded by the curves $y^2=4ax$, $y^2=4bx$, $xy=c^2$ and $xy=d^2$ is $\frac{1}{3}(d^2-c^2)\log\left(\frac{b}{a}\right); d>c, b>a.$ सिद्ध करें कि $\frac{1}{3}(d^2-c^2)\log\left(\frac{b}{a}\right); d>c, b>a.$ पॉजिटिव क्वाड्रंट में कर्व्स $y^2=4ax$, $y^2=4bx$, $xy=c^2$ और $xy=d^2$ से घिरा एरिया है	4	K1
b.	Find the volume of the solid bounded by the co-ordinate planes and the surface $\sqrt{\frac{x}{a}}+\sqrt{\frac{y}{b}}+\sqrt{\frac{z}{c}}=1.$ कोऑर्डिनेटप्लेन और सतह $\sqrt{\frac{x}{a}}+\sqrt{\frac{y}{b}}+\sqrt{\frac{z}{c}}=1$ से घिरे ठोस का वॉल्यूम पता करें।	4	K4

7. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Apply Green's theorem to evaluate $\int_C[(y-\sin x)\,dx+\cos x\,dy]$, where C is the triangle formed by $y=0,\ x=\frac{\pi}{2},\ y=\frac{2}{\pi}x.$ ग्रीन के प्रमेय को लागू करके $\int_C[(y-\sin x)\,dx+\cos x\,dy]$ का मूल्यांकन करें जहाँ C $y=0,\ x=\frac{\pi}{2},\ y=\frac{2}{\pi}x$ द्वारा बनाया गया त्रिभुज है।	5	K3
b.	Apply Gauss's divergence theorem to evaluate $\iint_S \bar{F}.\hat{n}\,ds$, where $\bar{F}=(2x^2-3z)\hat{i}-2xy\,\hat{j}-4x\hat{k}$ and S is the Surface bounded by the planes $x=0, y=0, z=0$ and $2x+2y+z=4.$ $\iint_S \bar{F}.\hat{n}\,ds$, जहाँ $\bar{F}=(2x^2-3z)\hat{i}-2xy\,\hat{j}-4x\hat{k}$ का मूल्यांकन करने के लिए	5	K5

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	गॉस के विचलन सिद्धांत को लागू करें $x = 0, y = 0, z = 0$ और $2x + 2y + z = 4$, S विमानों द्वारा घिरा सतह है।		

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AKTU - M1 - ALL UNITS - FROM 8 PAST PAPERS

Top 3 AKTU questions - all units

Engineering Mathematics-i. Done with the paper? These come back the most across the whole subject.

1 Change the order of integration and hence evaluate $\int_0^a\int_{\sqrt{ax}}^a\frac{y^2}{\sqrt{y^4-a^2x^2}}\,dy\,dx$.

- Unit 4 - 7 marks - asked 7 times - last 2026

2 If $z = x^4y^2 \sin^{-1}(\frac{x}{y}) + \log x - \log y$, show that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 6x^4y^2 \sin^{-1}(\frac{x}{y})$.

- Unit 2 - 7 marks - asked 7 times - last 2026

3 If $y = e^{\tan^{-1}x}$, prove that $(1 + x^2)y_{n+2} + [(2n+2)x - 1]y_{n+1} + n(n+1)y_n = 0$.

- Unit 2 - 10 marks - asked 8 times - last 2026

Practise all important questions >

Tick 'I can answer this' and track your prep.

EVERY UNIT - NOTES, IMP QS, PYQS, STRATEGY

Tap a link to open resources for that unit.

Unit 1 - Matrices	Notes ↗	IMP Qs ↗	PYQs ↗	Strategy ↗
Unit 2 - Differential Calculus-I	Notes ↗	IMP Qs ↗	PYQs ↗	Strategy ↗
Unit 3 - Differential Calculus-II	Notes ↗	IMP Qs ↗	PYQs ↗	Strategy ↗
Unit 4 - Multiple Integration		IMP Qs ↗	PYQs ↗	Strategy ↗
Unit 5 - Vector Calculus		IMP Qs ↗	PYQs ↗	Strategy ↗

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AKTU - M1

Paper Done.

See What Repeats InEngineering Mathematics-i

187
QUESTIONS ANALYSED

We analysed 8 AKTU papers - these questions keep coming back.

- **8 AKTU papers analysed | 187 questions | 2019-2026**

Must-do topics & questions
Questions that repeat in AKTU papers, ranked unit-wise. **Open >**

Predicted paper
A likely paper built from past AKTU papers. Practise it before the exam. **Open >**

Past papers (PYQ)
Filter by year, unit and topic. See which questions repeat. **Open >**

Subject strategy
Unit weightage, safest units and what to study first. **Open >**

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